Appendix

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Appendix A. Complete Mathematical Formulation

This appendix gives the complete derivation of the fixed and linear-parameter-varying equivalent-circuit models used in this study. It establishes the state recursion, state-of-charge feasibility conditions, physical parameter constraints, hierarchical identifiability, candidate-family dimensions, sparse-support closure, calibration-stage selection rule, and test-isolation properties. The trajectory and chemistry structure follows the open randomized-current lithium-ion degradation dataset used in the main study.

Appendix A.1. Complete notation and abbreviations

Table Appendix A.1: Complete mathematical notation and abbreviations used in the study.

Symbol	Meaning	Symbol	Meaning
Indices, sets, and identifiers			
s	Trajectory index	S	Total number of trajectories
k	Discrete-time sample index	N_s	Number of ordered samples in trajectory s
n	Summation index for charge increments	j	Polarization-branch index
J	Number of RC polarization branches	c	Chemistry index
c_s	Chemistry identity of trajectory s	C	Number of chemistries
ℓ_s	Cell identity of trajectory s	$\mathcal{L}_c$	Set of cells belonging to chemistry c
L	Total number of cells	f_s	Source-file identity of trajectory s
g_s	Segment index of trajectory s	UID_s	Unique trajectory identifier (c_s, ℓ_s, f_s, g_s)

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Table Appendix A.1 (continued)

Symbol	Meaning	Symbol	Meaning
m	Polynomial-basis index, $m = 1, \ldots, 6$	r	Transformed-parameter channel index
$\mathcal{R}$	Set $\{R_0, R_1, R_2, \tau_1, \Delta\tau\}$	M	Generic candidate-model index
$\mathcal{M}$	Candidate set $\{M_0, M_1, M_2, M_3, M_4\}$	g	Sparse coefficient-group index
$\mathcal{G}_M$	Group-sparsity collection for candidate M	i	Scalar coefficient or vector-component index
u	Optimization-iteration index	b	Mini-batch or diagnostic-bin index, according to context
B	Number of mini-batches in the gradient-span diagnostic		
Measured variables, states, and SOC reconstruction			
$t_{s,k}$	Elapsed time at sample k	$\Delta t_{s,k}$	Positive sampling interval $t_{s,k+1} - t_{s,k}$
$V_{s,k}$	Measured terminal voltage	$\widehat{V}_{s,k}$	Recursively predicted terminal voltage
$\widehat{V}_{s,k}^{(J)}$	Predicted voltage of a J-branch ECM	$e_{s,k}$	Signed voltage residual $V_{s,k} - \widehat{V}_{s,k}$
$I_{s,k}$	Cell current; positive current denotes discharge	$\lvert I_{s,k}\rvert$	Current magnitude
$T_{s,k}$	Measured cell temperature	$z_{s,k}$	Reconstructed state of charge
$z_{s,0}^{\mathrm{sel}}$	Selected initial SOC	Q_s	Capacity assigned to trajectory s, in ampere-hours
η_{cc}	Coulombic-efficiency factor	$\Delta q_{s,k}$	Signed charge increment over interval k, in ampere-hours
$q_{s,k}^{\mathrm{cum}}$	Cumulative signed charge	$d_{s,k}$	Cumulative SOC displacement
N_{segment}	Number of samples in a data segment	T_{duration}	Duration of a data segment
OCV support and initial-state quantities			
$z_j^{(c)}$	SOC coordinate of OCV-grid point j for chemistry c	$U_j^{(c)}$	OCV value at grid point j for chemistry c
M_c	Number of OCV-grid points for chemistry c	$\boldsymbol{z}_{\mathrm{grid}}^{(c)}$	Chemistry-specific vector of SOC support points
$\boldsymbol{U}_{\mathrm{grid}}^{(c)}$	Chemistry-specific vector of OCV values	$U_{\mathrm{oc}}^{(c)}(z)$	Chemistry-specific OCV function
$U_{\mathrm{oc},s,k}$	OCV supplied to the voltage model	$z_{\min}^{(c)}, z_{\max}^{(c)}$	Validated SOC-support limits for chemistry c
$U_{\min}^{(c)}, U_{\max}^{(c)}$	Validated OCV-voltage limits for chemistry c	$z_{s,k}^{\mathrm{OCV}}$	SOC clipped to the validated OCV support
$V_{s,0}^{\mathrm{OCV}}$	Initial-rest voltage clipped to the OCV voltage range	ϵ_{OCV}	Plateau-equivalence tolerance for inverse OCV evaluation
ϵ_{mon}	Numerical monotonicity tolerance	$\mathcal{Z}_{s,0}^{\mathrm{feas}}$	Charge-consistent feasible interval for initial SOC
$\mathcal{Z}_{s,0}^{\mathrm{OCV}}$	OCV-derived initial-SOC point or interval	$\mathcal{Z}_{s,0}^{\mathrm{con}}$	Intersection of OCV-derived and feasible initial-SOC intervals
$z_{s,0}^{\mathrm{OCV,lo}}, z_{s,0}^{\mathrm{OCV,hi}}$	Lower and upper limits of the OCV-derived interval	$z_{s,0}^{\mathrm{OCV,rep}}$	Representative OCV-derived initial SOC

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Table Appendix A.1 (continued)

Symbol	Meaning	Symbol	Meaning
$z_{s,0}^{\mathrm{con,lo}}, z_{s,0}^{\mathrm{con,hi}}$	Limits of the consistent initial-SOC interval	I_{rest}	Absolute-current threshold for rest detection
N_{rest}	Number of samples in the initial-rest window	$V_{s,\mathrm{rest}}$	Median voltage over the initial-rest window
Equivalent-circuit states and physical parameters			
R_0	Fixed ohmic resistance	R_j	Fixed polarization resistance of branch j
$R_{0,s,k}$	Sample-dependent ohmic resistance	$R_{j,s,k}$	Sample-dependent polarization resistance of branch j
$C_{j,s,k}$	Derived polarization capacitance of branch j	τ_j	Fixed polarization time constant of branch j
$\tau_{j,s,k}$	Sample-dependent polarization time constant	$\Delta\tau$	Positive fixed time-constant gap $\tau_2 - \tau_1$
$\Delta\tau_{s,k}$	Sample-dependent time-constant gap	$v_{j,s,k}$	Polarization-voltage state of branch j
$v_{p,s,0}^{\mathrm{tot}}$	Total initial polarization voltage	$\alpha_{j,s,k}$	Discrete RC decay coefficient
$I_{\max}$	Upper bound on current magnitude in the boundedness argument	$R_{\max}$	Upper bound on polarization resistance in the boundedness argument
$\Delta t_{\min}$	Positive lower bound on sampling interval	$\tau_{\max}$	Finite upper bound on time constant
$\overline{\alpha}$	Uniform upper bound on RC decay coefficients	$\psi_{R_0}, \psi_{R_1}, \psi_{R_2}$	Unconstrained transformed coordinates for fixed resistances
$\psi_{\tau_1}, \psi_{\Delta\tau}$	Unconstrained transformed coordinates for fixed time constants	$\Delta_{\mathrm{2RC}}^{\mathrm{cal}}$	Relative calibration-RMSE gain of 2RC over 1RC
LPV scheduler, basis, and transformed parameters			
$\boldsymbol{\chi}_{s,k}$	Raw SOC–temperature scheduling vector	$\boldsymbol{\chi}_{\min}^{\mathrm{tr}}, \boldsymbol{\chi}_{\max}^{\mathrm{tr}}$	Componentwise training-only scheduler limits
$\boldsymbol{\chi}_{\mathrm{mid}}$	Training-only scheduler midpoint	$\boldsymbol{\chi}_{\mathrm{half}}$	Training-only scheduler half-range
$\boldsymbol{\xi}_{s,k}$	Normalized SOC–temperature scheduler	$\xi_{z,s,k}$	Normalized SOC coordinate
$\xi_{T,s,k}$	Normalized temperature coordinate	$\boldsymbol{\phi}(\boldsymbol{\xi}_{s,k})$	Six-term polynomial basis vector
$\phi_m(\boldsymbol{\xi}_{s,k})$	Basis term m evaluated at the normalized scheduler	$\boldsymbol{\eta}_{s,k}$	Vector of transformed ECM parameter channels
$\eta_{R_i,s,k}$, $i \in \{0,1,2\}$	Transformed resistance channels	$\eta_{\tau_1,s,k}$	Transformed first-time-constant channel
$\eta_{\Delta\tau,s,k}$	Transformed time-constant-gap channel	$\widetilde{\eta}_{r,s,k}$	Clipped transformed predictor for channel r
$\eta_r^{\min}, \eta_r^{\max}$	Frozen lower and upper bounds for transformed channel r	$\oslash$	Componentwise division
Hierarchical coefficient representation			
$\beta_{r,m}^{G}$	Global coefficient for channel r and basis term m	$\beta_{r,c,m}^{C}$	Chemistry-level deviation coefficient
$\beta_{r,c,\ell,m}^{L}$	Cell-level deviation coefficient nested within chemistry	$n_{c,\ell}$	Training representation count for cell ℓ in chemistry c

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Table Appendix A.1 (continued)

Symbol	Meaning	Symbol	Meaning
N_c	Training representation count for chemistry c	N	Total training representation count in the hierarchy
$\theta_{c,\ell}$	Effective coefficient for cell ℓ in chemistry c	$\bar{\theta}$	Weighted grand mean of effective cell coefficients
$\bar{\theta}_c$	Weighted mean within chemistry c	$\boldsymbol{w}_C$	Vector of positive chemistry weights
$\boldsymbol{N}_C$	Orthonormal null-space basis for the chemistry constraint	$\boldsymbol{\gamma}^C$	Reduced chemistry-deviation coordinates
$\boldsymbol{w}_{L,c}$	Vector of positive cell weights in chemistry c	$\boldsymbol{N}_{L,c}$	Orthonormal null-space basis for the cell constraint
$\boldsymbol{\gamma}_c^L$	Reduced cell-deviation coordinates	$\boldsymbol{\beta}^{\text{full}}$	Complete hierarchical coefficient vector
$\boldsymbol{\beta}^{\text{red}}$	Identifiable reduced coefficient vector	$\boldsymbol{A}_{\text{id}}$	Frozen expansion from reduced to complete coefficients
p_{one}	Reduced dimension for one channel and one basis term	$p_{\text{full}}, p_{\text{red}}$	Complete and reduced hierarchy dimensions
Candidate families, masks, and support			
M_0	Fixed global 2RC candidate	M_1	Global dense SOC–temperature LPV candidate
M_2	Global sparse SOC–temperature LPV candidate	M_3	Hierarchical dense SOC–temperature LPV candidate
M_4	Hierarchical sparse SOC–temperature LPV candidate	$\mathcal{A}_M$	Coefficient space enabled by candidate M
$\boldsymbol{m}_M^{\text{cand}}$	Binary reduced-coordinate mask for candidate M	$\boldsymbol{\beta}_M$	Candidate-enabled reduced coefficient vector
$\odot$	Elementwise multiplication	$p_M, p_M^{\text{full}}, p_M^{\text{red}}$	Candidate coefficient dimensions
$\eta_{R_i}^{(0)}, \eta_{\tau_1}^{(0)}, \eta_{\Delta\tau}^{(0)}$	Initial transformed values from the fixed 2RC benchmark	β_i	Scalar reduced-coordinate coefficient
$\boldsymbol{\beta}_g$	Coefficient subvector belonging to sparse group g	$\lvert g\rvert$	Number of coefficients in sparse group g
$\epsilon_{\text{support}}$	Coefficient/group activity threshold	$\mathcal{A}_{\text{raw}}$	Support obtained directly by thresholding
$\text{cl}(\mathcal{A}_{\text{raw}})$	Smallest heredity-valid superset of the raw support	$\boldsymbol{m}^{\text{closed}}$	Binary mask of the heredity-closed support
Losses, regularization, and optimization			
$\rho_\delta(r)$	Huber loss with transition scale δ	$\ell_{\text{PH}}(e;\delta)$	Pseudo-Huber loss
δ	Huber or pseudo-Huber transition scale	$m_{s,k}^{\text{loss}}$	Binary fitting-loss mask
N_{loss}	Number of samples included in the fitting loss	$\mathcal{L}_{\text{data}}$	Normalized pseudo-Huber data-loss term
$\mathcal{L}_{\text{ridge},M}$	Candidate-specific ridge term	λ_{ridge}	Ridge-regularization weight
$\mathcal{P}_M(\boldsymbol{\beta})$	Group-lasso penalty for sparse candidate M	$d_{s,k,r}^{-}, d_{s,k,r}^{+}$	Lower- and upper-bound excursions of transformed predictors
$\mathcal{B}_{\text{phys}}$	Physical-range excursion penalty	λ_{phys}	Weight of the physical-range penalty
λ_M	Group-sparsity penalty weight for candidate M	$\mathcal{J}_M$	Complete candidate-training objective

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Table Appendix A.1 (continued)

Symbol	**Meaning**	**Symbol**	**Meaning**
$\mathcal{J}_{\mathrm{refit}}$	Training-only selected-support refit objective	α_{Adam}	Adam learning rate
$\beta_{1,\mathrm{Adam}}$	Adam first-moment decay coefficient	$\beta_{2,\mathrm{Adam}}$	Adam second-moment decay coefficient
ϵ_{Adam}	Adam numerical-stability constant	B_{traj}	Number of trajectories per optimization batch
$\boldsymbol{g}^{(u)}$	Raw gradient at optimization iteration u	$\widetilde{\boldsymbol{g}}^{(u)}$	Norm-clipped gradient
Calibration selection, freezing, and metrics			
$\mathcal{S}_{\mathrm{cal}}$	Set of calibration trajectories with valid evaluated samples	$\mathcal{K}_s^{\mathrm{eval}}$	Evaluated sample-index set for trajectory s
N_s^{eval}	Number of evaluated samples in trajectory s	$\mathrm{RMSE}_{j,s}^{\mathrm{cal}}$	Calibration RMSE of job j on trajectory s
E_j^{cal}	Trajectory-weighted calibration score for job j	$E_{\mathrm{min}}^{\mathrm{cal}}$	Minimum trajectory-weighted calibration score
$\epsilon_{\mathrm{cal,eq}}$	Relative calibration-equivalence tolerance	$\mathcal{A}_{\mathrm{cal}}$	Calibration-equivalent admissible job pool
$M^\star$	Selected candidate	$\lambda^\star$	Selected regularization value
$\boldsymbol{\beta}^\star$	Frozen selected coefficient vector	$\mathcal{F}$	Set of all objects frozen before test access
$m_{s,k}^{\mathrm{eval}}$	Binary evaluation mask	$\mathcal{E}$	Nonempty pooled set of evaluated trajectory–sample pairs
$\mathcal{S}_{\mathrm{eval}}$	Set of trajectories represented in $\mathcal{E}$	$\mathrm{RMSE}_{\mathrm{pool}}$	Pooled root-mean-square voltage error
$\mathrm{MAE}_{\mathrm{pool}}$	Pooled mean absolute voltage error	$\mathrm{Bias}_{\mathrm{pool}}$	Pooled mean signed residual
RMSE_s	RMSE for trajectory s	$\mathrm{RMSE}_{\mathrm{TW}}$	Trajectory-weighted RMSE
Residual and scheduler-support diagnostics			
$x_{s,k}$	Generic operating variable used in residual diagnostics	$\overline{e}, \overline{x}$	Sample means of residual and diagnostic variable
$\rho(e,x)$	Sample-weighted Pearson correlation	$\mathcal{B}_b$	One-dimensional diagnostic bin
$\overline{e}_b$	Mean residual in bin b	$\mathcal{Z}_a$	SOC bin a
$\mathcal{T}_b$	Temperature bin b	$n_{a,b}$	Sample count in joint SOC–temperature bin (a,b)
$\overline{e}_{a,b}$	Mean residual in joint bin (a,b)	$\mathcal{E}_{\mathrm{tr}}$	Training sample set used for support-density calculation
Ξ_a	Normalized SOC-support bin	Θ_b	Normalized temperature-support bin
$n_{a,b}^{\mathrm{tr}}$	Training support count in normalized scheduler bin (a,b)	$\mathbb{I}\{\cdot\}$	Indicator function
Gradient-span diagnostic			
$p_{\mathcal{S}}$	Number of reduced coefficients in the selected support	$\boldsymbol{g}_b$	Selected-support data-loss gradient for mini-batch b
$\mathcal{L}_{\mathrm{data}}^{(b)}$	Data loss of mini-batch b	$\boldsymbol{G}$	Matrix of mini-batch gradient row vectors
$\overline{\boldsymbol{g}}$	Mean mini-batch gradient	$\boldsymbol{G}_c$	Centered mini-batch gradient matrix

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Table Appendix A.1 (continued)

Symbol	**Meaning**	**Symbol**	**Meaning**
$\mathbf{1}_B$	Length-B vector of ones	$\boldsymbol{H}_B$	Mini-batch centering matrix
$\boldsymbol{I}_B$	Identity matrix of order B	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$	Matrices in the singular-value decomposition of $\boldsymbol{G}_c$
$\sigma_i(\boldsymbol{G}_c)$	The i-th singular value of $\boldsymbol{G}_c$	ϵ_{mach}	Machine floating-point precision
r_{grad}	Practical numerical rank of the centered gradient span	$\#\{\cdot\}$	Cardinality of a condition-defined set
Abbreviations			
ECM	Equivalent-circuit model	LPV	Linear-parameter-varying model
1RC	Equivalent-circuit model with one RC polarization branch	2RC	Equivalent-circuit model with two RC polarization branches
SOC	State of charge	OCV	Open-circuit voltage
BMS	Battery-management system	RMSE	Root-mean-square error
MAE	Mean absolute error	LFP	Lithium iron phosphate
NCA	Lithium nickel cobalt aluminium oxide	NMC	Lithium nickel manganese cobalt oxide
LMO	Lithium manganese oxide	LTO	Lithium titanate
RLS	Recursive least squares	LASSO	Least absolute shrinkage and selection operator
GPR	Gaussian-process regression	SOH	State of health
EKF	Extended Kalman filter	QLPV	Quasi-linear-parameter-varying model

Appendix A.2. Trajectory notation and current convention

Complete trajectories are indexed by $s \in \{1, \ldots, S\}$. Trajectory s contains N_s ordered samples indexed by

$$k = 0, 1, \ldots, N_s - 1.$$

The elapsed time, measured terminal voltage, current, temperature, and reconstructed state of charge are denoted by

$$t_{s,k}, \qquad V_{s,k}, \qquad I_{s,k}, \qquad T_{s,k}, \qquad z_{s,k},$$

respectively. The sampling interval is

$$\Delta t_{s,k} = t_{s,k+1} - t_{s,k}, \qquad \Delta t_{s,k} > 0, \qquad k = 0, \ldots, N_s - 2. \quad \text{(Appendix A.1)}$$

Positive current denotes discharge:

$$I_{s,k} > 0 \quad \text{denotes discharge.} \qquad \text{(Appendix A.2)}$$

Negative current denotes charge, whereas zero current denotes rest under the adopted sign convention.

Appendix A.3. State-of-charge reconstruction

The signed charge transferred over the interval $[t_{s,k}, t_{s,k+1}]$ is evaluated by trapezoidal integration:

$$\Delta q_{s,k} = \frac{\Delta t_{s,k}\,(I_{s,k} + I_{s,k+1})}{2 \times 3600}, \qquad k = 0, \ldots, N_s - 2, \qquad \text{(Appendix A.3)}$$

where $\Delta q_{s,k}$ is expressed in ampere-hours. The cumulative signed charge is

$$q^{\mathrm{cum}}_{s,0} = 0, \qquad q^{\mathrm{cum}}_{s,k} = \sum_{n=0}^{k-1} \Delta q_{s,n}, \qquad k \geq 1. \qquad \text{(Appendix A.4)}$$

Let $Q_s > 0$ denote the capacity assigned to trajectory s, and let η_{cc} denote the Coulombic-efficiency factor. The implemented analysis uses

$$\eta_{\mathrm{cc}} = 1. \qquad \text{(Appendix A.5)}$$

Define the cumulative SOC displacement as

$$d_{s,k} = \frac{\eta_{\mathrm{cc}} q^{\mathrm{cum}}_{s,k}}{Q_s}. \qquad \text{(Appendix A.6)}$$

The reconstructed SOC is then

$$z_{s,k} = z^{\mathrm{sel}}_{s,0} - d_{s,k}, \qquad \text{(Appendix A.7)}$$

or, equivalently,

$$z_{s,k+1} = z_{s,k} - \frac{\eta_{\mathrm{cc}} \Delta t_{s,k}\,(I_{s,k} + I_{s,k+1})}{7200 Q_s}. \qquad \text{(Appendix A.8)}$$

The numerical SOC-validity interval used during data preparation is

$$-0.02 \leq z_{s,k} \leq 1.02. \qquad \text{(Appendix A.9)}$$

This tolerance permits small reconstruction deviations around the physical interval $[0, 1]$ while excluding inadmissible SOC values.

Equation (Appendix A.8) is consistent with the adopted current convention. During discharge, $I_{s,k} > 0$, so the reconstructed SOC decreases. During charge, $I_{s,k} < 0$, so the reconstructed SOC increases.

Appendix A.4. Charge-consistent initial-SOC interval

For chemistry c_s, let the validated OCV support be

$$z_{\min}^{(c_s)} \le z \le z_{\max}^{(c_s)}. \tag{Appendix A.10}$$

Because

$$z_{s,k} = z_{s,0}^{\text{sel}} - d_{s,k},$$

requiring every reconstructed SOC value to remain within the validated support gives

$$z_{\min}^{(c_s)} \le z_{s,0}^{\text{sel}} - d_{s,k} \le z_{\max}^{(c_s)} \qquad \text{for all } k. \tag{Appendix A.11}$$

Rearranging the two inequalities yields

$$z_{\min}^{(c_s)} + d_{s,k} \le z_{s,0}^{\text{sel}} \le z_{\max}^{(c_s)} + d_{s,k}. \tag{Appendix A.12}$$

The initial SOC must therefore belong to

$$\mathcal{Z}_{s,0}^{\text{feas}} = \bigcap_{k=0}^{N_s-1} \left[z_{\min}^{(c_s)} + d_{s,k},\, z_{\max}^{(c_s)} + d_{s,k} \right]. \tag{Appendix A.13}$$

The intersection is

$$\mathcal{Z}_{s,0}^{\text{feas}} = \left[z_{\min}^{(c_s)} + \max_k d_{s,k},\, z_{\max}^{(c_s)} + \min_k d_{s,k} \right]. \tag{Appendix A.14}$$

Proof.. For each k, the lower bound on $z_{s,0}^{\text{sel}}$ is $z_{\min}^{(c_s)} + d_{s,k}$. Satisfying all lower bounds is equivalent to satisfying their maximum. Similarly, satisfying all upper bounds is equivalent to remaining below their minimum. Hence, Eq. (Appendix A.14) follows. The interval is nonempty if and only if

$$z_{\min}^{(c_s)} + \max_k d_{s,k} \le z_{\max}^{(c_s)} + \min_k d_{s,k}. \tag{Appendix A.15}$$

□

Appendix A.5. Initial-rest detection and voltage statistic

An initial rest window is detected when

$$|I_{s,k}| \le I_{\text{rest}}, \qquad k = 0, \ldots, N_{\text{rest}} - 1, \tag{Appendix A.16}$$

with

$$I_{\text{rest}} = 0.05 \text{ A}, \qquad N_{\text{rest}} = 5. \tag{Appendix A.17}$$

When this condition holds, the initial-rest voltage is

$$V_{s,\text{rest}} = \text{median}\left(V_{s,0}, \ldots, V_{s,N_{\text{rest}}-1}\right). \tag{Appendix A.18}$$

A reported initial SOC is used directly only when it is finite, lies within the admissible SOC interval, and is compatible with the charge-consistent interval in Eq. (Appendix A.14). Otherwise, the chemistry-specific inverse OCV rule below is applied only when the initial-rest condition is satisfied. A trajectory without either a valid reported initial SOC or a valid initial-rest estimate is excluded from parameter-fitting evidence.

Appendix A.6. OCV-derived initial-SOC interval

Chemistry-specific OCV construction and inversion follow the established role of OCV–SOC characterization in equivalent-circuit modelling [**?** **?**]. Let the chemistry-specific OCV relation be represented by

$$\left\{\left(z_j^{(c)}, U_j^{(c)}\right)\right\}_{j=1}^{M_c}. \tag{Appendix A.19}$$

The forward OCV map is

$$U_{\text{oc}}^{(c)}(z) = \text{interp}\left(z; \boldsymbol{z}_{\text{grid}}^{(c)}, \boldsymbol{U}_{\text{grid}}^{(c)}\right). \tag{Appendix A.20}$$

Before forward interpolation, SOC is restricted to the validated chemistry-specific support:

$$z_{s,k}^{\text{OCV}} = \text{clip}\left(z_{s,k}; z_{\min}^{(c_s)}, z_{\max}^{(c_s)}\right), \tag{Appendix A.21}$$

where

$$\text{clip}(x; a, b) = \min(\max(x, a), b). \tag{Appendix A.22}$$

The OCV input to the ECM is

$$U_{\text{oc},s,k} = U_{\text{oc}}^{(c_s)}\left(z_{s,k}^{\text{OCV}}\right). \tag{Appendix A.23}$$

Nearly flat OCV regions are treated as plateau-equivalent. The voltage tolerance is

$$\epsilon_{\text{OCV}} = \max\left(10^{-8} \text{ V}, 10\epsilon_{\text{mon}}\right), \qquad \epsilon_{\text{mon}} = 10^{-9} \text{ V}. \tag{Appendix A.24}$$

Let

$$U_{\min}^{(c)} = \min_{1\le j\le M_c} U_j^{(c)}, \qquad U_{\max}^{(c)} = \max_{1\le j\le M_c} U_j^{(c)}. \tag{Appendix A.25}$$

Before inverse OCV evaluation, the rest voltage is restricted to the validated chemistry-specific voltage range:

$$V_{s,0}^{\mathrm{OCV}} = \mathrm{clip}\left(V_{s,\mathrm{rest}}; U_{\min}^{(c_s)}, U_{\max}^{(c_s)}\right). \tag{Appendix A.26}$$

The plateau-aware set-valued inverse is

$$\mathcal{Z}_{s,0}^{\mathrm{OCV}} = \left\{ z \in \left[z_{\min}^{(c_s)}, z_{\max}^{(c_s)}\right] : \left|U_{\mathrm{oc}}^{(c_s)}(z) - V_{s,0}^{\mathrm{OCV}}\right| \le \epsilon_{\mathrm{OCV}} \right\}. \tag{Appendix A.27}$$

For a continuous, validated monotone OCV relation, this set is a single point or a closed SOC interval:

$$\mathcal{Z}_{s,0}^{\mathrm{OCV}} = \left[z_{s,0}^{\mathrm{OCV,lo}}, z_{s,0}^{\mathrm{OCV,hi}}\right]. \tag{Appendix A.28}$$

A deterministic representative of this point or interval is defined by

$$z_{s,0}^{\mathrm{OCV,rep}} = \frac{z_{s,0}^{\mathrm{OCV,lo}} + z_{s,0}^{\mathrm{OCV,hi}}}{2}. \tag{Appendix A.29}$$

The OCV-derived and charge-consistent evidence are combined through

$$\mathcal{Z}_{s,0}^{\mathrm{con}} = \mathcal{Z}_{s,0}^{\mathrm{OCV}} \cap \mathcal{Z}_{s,0}^{\mathrm{feas}}. \tag{Appendix A.30}$$

If this intersection is nonempty, a representative OCV estimate $z_{s,0}^{\mathrm{OCV,rep}}$ is projected onto the consistent interval:

$$z_{s,0}^{\mathrm{sel}} = \mathrm{clip}\left(z_{s,0}^{\mathrm{OCV,rep}}; z_{s,0}^{\mathrm{con,lo}}, z_{s,0}^{\mathrm{con,hi}}\right). \tag{Appendix A.31}$$

If the intersection is empty, the trajectory is not admitted as parameter-fitting evidence.

The OCV relation, capacity assignment, and reconstructed SOC are fixed before candidate fitting. They are not free parameters of the ECM or LPV optimizer.

Appendix A.7. Continuous-time RC branch and exact discrete recursion

For polarization branch j, the continuous-time equation is

$$C_{j,s,k}\frac{\mathrm{d}v_j(t)}{\mathrm{d}t} = I_{s,k} - \frac{v_j(t)}{R_{j,s,k}}, \qquad t \in [t_{s,k}, t_{s,k+1}), \tag{Appendix A.32}$$

where the current and parameter values evaluated at sample k are held constant over the interval. Define

$$\tau_{j,s,k} = R_{j,s,k}C_{j,s,k}. \tag{Appendix A.33}$$

The capacitance is therefore a derived quantity,

$$C_{j,s,k} = \frac{\tau_{j,s,k}}{R_{j,s,k}}, \tag{Appendix A.34}$$

rather than an independently fitted channel. Equation (Appendix A.32) becomes

$$\frac{\mathrm{d}v_j(t)}{\mathrm{d}t} + \frac{1}{\tau_{j,s,k}}v_j(t) = \frac{R_{j,s,k}}{\tau_{j,s,k}}I_{s,k}. \tag{Appendix A.35}$$

Applying the integrating factor $\exp(t/\tau_{j,s,k})$ gives

$$\begin{aligned} v_{j,s,k+1} = {} & \exp\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right) v_{j,s,k} \\ & + R_{j,s,k}\left[1 - \exp\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right)\right] I_{s,k}. \end{aligned} \tag{Appendix A.36}$$

Define

$$\alpha_{j,s,k} = \exp\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right). \tag{Appendix A.37}$$

The recursive update is

$$v_{j,s,k+1} = \alpha_{j,s,k}v_{j,s,k} + R_{j,s,k}\left(1 - \alpha_{j,s,k}\right) I_{s,k}. \tag{Appendix A.38}$$

For numerical accuracy when $\Delta t_{s,k}/\tau_{j,s,k}$ is small,

$$1 - \alpha_{j,s,k} = -\operatorname{expm1}\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right). \tag{Appendix A.39}$$

Appendix A.8. Fixed 1RC and 2RC voltage models

The following 1RC and 2RC structures are standard low-order Thévenin representations used in lithium-ion battery modelling. For a J-branch equivalent-circuit model,

$$\widehat{V}_{s,k}^{(J)} = U_{\mathrm{oc},s,k} - R_{0,s,k} I_{s,k} - \sum_{j=1}^{J} v_{j,s,k}. \qquad \text{(Appendix A.40)}$$

For the fixed 1RC model,

$$\widehat{V}_{s,k}^{\mathrm{1RC}} = U_{\mathrm{oc},s,k} - R_0 I_{s,k} - v_{1,s,k}, \qquad \text{(Appendix A.41)}$$

with

$$v_{1,s,k+1} = \alpha_{1,s,k} v_{1,s,k} + R_1 \left(1 - \alpha_{1,s,k}\right) I_{s,k}. \qquad \text{(Appendix A.42)}$$

For the fixed 2RC model,

$$\widehat{V}_{s,k}^{\mathrm{2RC}} = U_{\mathrm{oc},s,k} - R_0 I_{s,k} - v_{1,s,k} - v_{2,s,k}, \qquad \text{(Appendix A.43)}$$

with Eq. (Appendix A.38) applied for $j \in \{1, 2\}$.

For the fixed models, positivity and time-constant ordering are enforced through

$$R_0 = \exp(\psi_{R_0}), \qquad R_1 = \exp(\psi_{R_1}), \qquad \tau_1 = \exp(\psi_{\tau_1}), \quad \text{(Appendix A.44)}$$

and, for the 2RC model,

$$R_2 = \exp(\psi_{R_2}), \qquad \Delta\tau = \exp(\psi_{\Delta\tau}), \qquad \tau_2 = \tau_1 + \Delta\tau. \quad \text{(Appendix A.45)}$$

The fixed-model optimizer uses the same physical intervals later stated in Eqs. (Appendix A.77)–(Appendix A.79).

The fixed-order comparison uses the Huber loss

$$\rho_\delta(r) = \begin{cases} \dfrac{1}{2} r^2, & |r| \le \delta, \\ \delta|r| - \dfrac{1}{2}\delta^2, & |r| > \delta, \end{cases} \qquad \delta = 0.020 \text{ V}. \qquad \text{(Appendix A.46)}$$

Let

$$\Delta_{\mathrm{2RC}}^{\mathrm{cal}} = \frac{\mathrm{RMSE}_{\mathrm{1RC}}^{\mathrm{cal}} - \mathrm{RMSE}_{\mathrm{2RC}}^{\mathrm{cal}}}{\mathrm{RMSE}_{\mathrm{1RC}}^{\mathrm{cal}}}. \qquad \text{(Appendix A.47)}$$

The 2RC structure is selected as the fixed baseline when

$$\Delta_{\mathrm{2RC}}^{\mathrm{cal}} \ge 0.005. \qquad \text{(Appendix A.48)}$$

This rule requires at least a 0.5% relative calibration-RMSE reduction before accepting the additional polarization branch.

Appendix A.9. Stability and boundedness of the RC recursion

Because

$$\Delta t_{s,k} > 0 \qquad \text{and} \qquad \tau_{j,s,k} > 0,$$

$$0 < \alpha_{j,s,k} < 1. \tag{Appendix A.49}$$

Proof.. For any $x > 0$, $0 < \exp(-x) < 1$. Taking $x = \Delta t_{s,k}/\tau_{j,s,k} > 0$ gives Eq. (Appendix A.49). □

Suppose that

$$|I_{s,k}| \leq I_{\max}, \qquad 0 < R_{j,s,k} \leq R_{\max}.$$

Then

$$|v_{j,s,k+1}| \leq \alpha_{j,s,k}|v_{j,s,k}| + (1 - \alpha_{j,s,k})\, R_{\max} I_{\max}. \tag{Appendix A.50}$$

It follows inductively that

$$|v_{j,s,k}| \leq \max\left(|v_{j,s,0}|, R_{\max} I_{\max}\right). \tag{Appendix A.51}$$

Thus, bounded current and bounded positive resistance imply bounded polarization states.

If

$$\Delta t_{s,k} \geq \Delta t_{\min} > 0, \qquad \tau_{j,s,k} \leq \tau_{\max} < \infty,$$

then

$$\alpha_{j,s,k} \leq \exp\left(-\frac{\Delta t_{\min}}{\tau_{\max}}\right) = \overline{\alpha} < 1, \tag{Appendix A.52}$$

which gives uniform exponential contraction of the homogeneous state recursion.

Appendix A.10. Polarization-state initialization

For a trajectory that satisfies Eq. (Appendix A.16), all polarization states are initialized to zero:

$$v_{j,s,0} = 0, \qquad j = 1, \ldots, J. \tag{Appendix A.53}$$

Otherwise, the total initial polarization voltage is

$$v_{p,s,0}^{\text{tot}} = U_{\text{oc},s,0} - R_{0,s,0} I_{s,0} - V_{s,0}. \tag{Appendix A.54}$$

For the 1RC model, the single initial polarization state is

$$v_{1,s,0} = v_{p,s,0}^{\mathrm{tot}}. \qquad \text{(Appendix A.55)}$$

For the 2RC model, the deterministic resistance-weighted allocation is

$$v_{1,s,0} = \frac{R_{1,s,0}}{R_{1,s,0} + R_{2,s,0}} v_{p,s,0}^{\mathrm{tot}}, \qquad \text{(Appendix A.56)}$$

and

$$v_{2,s,0} = \frac{R_{2,s,0}}{R_{1,s,0} + R_{2,s,0}} v_{p,s,0}^{\mathrm{tot}}. \qquad \text{(Appendix A.57)}$$

Since $R_{1,s,0} > 0$ and $R_{2,s,0} > 0$,

$$v_{1,s,0} + v_{2,s,0} = v_{p,s,0}^{\mathrm{tot}}. \qquad \text{(Appendix A.58)}$$

Therefore,

$$\begin{aligned} \widehat{V}_{s,0} &= U_{\mathrm{oc},s,0} - R_{0,s,0} I_{s,0} - v_{p,s,0}^{\mathrm{tot}} \\ &= V_{s,0}. \end{aligned} \qquad \text{(Appendix A.59)}$$

The decomposition between the two RC branches is not uniquely identifiable from a single voltage observation. The resistance-weighted allocation is therefore a reproducible initialization policy, not a proof of the true physical branch states. The first sample is excluded from the fitting loss.

Appendix A.11. Training-only SOC–temperature scheduler

State-dependent and LPV battery formulations motivate scheduling the electrical representation with measured or reconstructed operating variables [**?** 4**?** , 5, 6]. The implemented LPV scheduling vector contains reconstructed SOC and measured temperature only:

$$\boldsymbol{\chi}_{s,k} = \begin{bmatrix} z_{s,k} \\ T_{s,k} \end{bmatrix}. \qquad \text{(Appendix A.60)}$$

Signed current, current magnitude, current direction, and elapsed time are diagnostic variables rather than continuous scheduling coordinates. Chemistry and cell identity enter the hierarchical coefficient representation.

Let $\boldsymbol{\chi}_{\min}^{\mathrm{tr}}$ and $\boldsymbol{\chi}_{\max}^{\mathrm{tr}}$ denote the componentwise limits calculated from the training partition. Define

$$\boldsymbol{\chi}_{\mathrm{mid}} = \frac{\boldsymbol{\chi}_{\max}^{\mathrm{tr}} + \boldsymbol{\chi}_{\min}^{\mathrm{tr}}}{2}, \qquad \text{(Appendix A.61)}$$

and

$$\boldsymbol{\chi}_{\text{half}} = \frac{\boldsymbol{\chi}^{\text{tr}}_{\max} - \boldsymbol{\chi}^{\text{tr}}_{\min}}{2}. \qquad \text{(Appendix A.62)}$$

Each scheduler coordinate must satisfy

$$\chi_{\text{half},i} > 0. \qquad \text{(Appendix A.63)}$$

A coordinate with zero half-range cannot identify parameter variation and must be removed from the scheduling dictionary or assigned a predeclared positive scale. Both implemented scheduler coordinates have strictly positive training half-ranges. The normalized scheduler is

$$\boldsymbol{\xi}_{s,k} = (\boldsymbol{\chi}_{s,k} - \boldsymbol{\chi}_{\text{mid}}) \oslash \boldsymbol{\chi}_{\text{half}} = \begin{bmatrix} \xi_{z,s,k} \\ \xi_{T,s,k} \end{bmatrix}, \qquad \text{(Appendix A.64)}$$

where $\oslash$ denotes componentwise division.

For the implemented training partition,

$$z^{\text{tr}}_{\min} = 0, \qquad z^{\text{tr}}_{\max} = 1, \qquad \text{(Appendix A.65)}$$

and

$$T^{\text{tr}}_{\min} = 23.5929\,^{\circ}\text{C}, \qquad T^{\text{tr}}_{\max} = 57.9206\,^{\circ}\text{C}. \qquad \text{(Appendix A.66)}$$

Thus,

$$\xi_{z,s,k} = \frac{z_{s,k} - 0.5}{0.5}, \qquad \text{(Appendix A.67)}$$

and

$$\xi_{T,s,k} = \frac{T_{s,k} - 40.75675}{17.16385}. \qquad \text{(Appendix A.68)}$$

The normalized scheduler is not clipped to $[-1, 1]^2$. Held-out samples outside the training range therefore remain outside the normalized training domain.

The frozen basis is

$$\boldsymbol{\phi}(\boldsymbol{\xi}_{s,k}) = \begin{bmatrix} 1 \\ \xi_{z,s,k} \\ \xi_{T,s,k} \\ \xi^2_{z,s,k} \\ \xi^2_{T,s,k} \\ \xi_{z,s,k}\xi_{T,s,k} \end{bmatrix} \in \mathbb{R}^6. \qquad \text{(Appendix A.69)}$$

Appendix A.12. Transformed physical parameters

The transformed representation is consistent with battery ECM studies in which resistance and polarization parameters vary with SOC, temperature, chemistry, and operating condition. All logarithmic and exponential transformations are applied to numerical parameter values expressed in the stated base units (Ω for resistance and seconds for time), so the transformed coordinates are dimensionless numerical quantities. The transformed channels are

$$\boldsymbol{\eta}_{s,k} = \begin{bmatrix} \eta_{R_0,s,k} \\ \eta_{R_1,s,k} \\ \eta_{R_2,s,k} \\ \eta_{\tau_1,s,k} \\ \eta_{\Delta\tau,s,k} \end{bmatrix}. \tag{Appendix A.70}$$

For channel r,

$$\widetilde{\eta}_{r,s,k} = \mathrm{clip}\left(\eta_{r,s,k}; \eta_r^{\min}, \eta_r^{\max}\right). \tag{Appendix A.71}$$

The physical parameters used by the simulator are

$$R_{0,s,k} = \exp\left(\widetilde{\eta}_{R_0,s,k}\right), \tag{Appendix A.72}$$

$$R_{1,s,k} = \exp\left(\widetilde{\eta}_{R_1,s,k}\right), \qquad R_{2,s,k} = \exp\left(\widetilde{\eta}_{R_2,s,k}\right), \tag{Appendix A.73}$$

$$\tau_{1,s,k} = \exp\left(\widetilde{\eta}_{\tau_1,s,k}\right), \tag{Appendix A.74}$$

$$\Delta\tau_{s,k} = \exp\left(\widetilde{\eta}_{\Delta\tau,s,k}\right), \tag{Appendix A.75}$$

and

$$\tau_{2,s,k} = \tau_{1,s,k} + \Delta\tau_{s,k}. \tag{Appendix A.76}$$

The frozen physical intervals are

$$10^{-5}\ \Omega \leq R_{0,s,k}, R_{1,s,k}, R_{2,s,k} \leq 1\ \Omega, \tag{Appendix A.77}$$

$$10^{-2}\ \mathrm{s} \leq \tau_{1,s,k} \leq 2 \times 10^{3}\ \mathrm{s}, \tag{Appendix A.78}$$

and

$$10^{-2}\ \mathrm{s} \leq \Delta\tau_{s,k} \leq 2 \times 10^{4}\ \mathrm{s}. \tag{Appendix A.79}$$

Positivity and ordering proof.. For any real x, $\exp(x) > 0$. Hence,

$$R_{0,s,k} > 0, \quad R_{1,s,k} > 0, \quad R_{2,s,k} > 0, \quad \tau_{1,s,k} > 0, \quad \Delta\tau_{s,k} > 0.$$

Since

$$\tau_{2,s,k} = \tau_{1,s,k} + \Delta\tau_{s,k},$$

$$0 < \tau_{1,s,k} < \tau_{2,s,k}. \qquad \text{(Appendix A.80)}$$

The transformation therefore enforces positive physical parameters and removes branch-label exchange between unordered time constants. □

The simulator uses the clipped predictors $\widetilde{\eta}_{r,s,k}$, while the objective penalizes excursions of the unclipped predictors $\eta_{r,s,k}$.

Appendix A.13. Hierarchical SOC–temperature parameter map

Let

$$\mathcal{R} = \{R_0, R_1, R_2, \tau_1, \Delta\tau\}.$$

For chemistry c_s, cell ℓ_s, and basis term m,

$$\eta_{r,s,k} = \sum_{m=1}^{6} \left(\beta^G_{r,m} + \beta^C_{r,c_s,m} + \beta^L_{r,c_s,\ell_s,m}\right) \phi_m(\boldsymbol{\xi}_{s,k}). \qquad \text{(Appendix A.81)}$$

Here, $\beta^G_{r,m}$ is the global coefficient, $\beta^C_{r,c,m}$ is the chemistry-level deviation, and $\beta^L_{r,c,\ell,m}$ is the cell-level deviation nested within chemistry c.

Let $n_{c,\ell} > 0$ denote the training representation count for cell ℓ in chemistry c, and define

$$N_c = \sum_{\ell \in \mathcal{L}_c} n_{c,\ell}, \qquad N = \sum_{c=1}^{C} N_c. \qquad \text{(Appendix A.82)}$$

Weighted centring removes the nonuniqueness between global and nested deviation terms. Related structured and hierarchical sparsity principles are discussed in [7, 6]. For each transformed channel and basis term, impose

$$\sum_{c=1}^{C} N_c \beta^C_{r,c,m} = 0, \qquad \text{(Appendix A.83)}$$

and

$$\sum_{\ell \in \mathcal{L}_c} n_{c,\ell} \beta^L_{r,c,\ell,m} = 0. \qquad \text{(Appendix A.84)}$$

Appendix A.14. Uniqueness of the hierarchical decomposition

For fixed transformed channel and basis term, suppress r, m and define the effective cell coefficient as

$$\theta_{c,\ell} = \beta^G + \beta_c^C + \beta_{c,\ell}^L. \tag{Appendix A.85}$$

Proposition A.1.. Under positive training weights and the weighted zero-sum constraints, the global, chemistry-level, and cell-level decomposition is unique.

Proof.. The weighted grand mean is

$$\begin{aligned}\overline{\theta} &= \frac{1}{N}\sum_{c=1}^{C}\sum_{\ell\in\mathcal{L}_c} n_{c,\ell}\theta_{c,\ell} \\ &= \beta^G + \frac{1}{N}\sum_{c=1}^{C} N_c\beta_c^C + \frac{1}{N}\sum_{c=1}^{C}\sum_{\ell\in\mathcal{L}_c} n_{c,\ell}\beta_{c,\ell}^L. \end{aligned}\tag{Appendix A.86}$$

The final two terms vanish by Eqs. (Appendix A.83) and (Appendix A.84). Therefore,

$$\beta^G = \overline{\theta}. \tag{Appendix A.87}$$

Within chemistry c,

$$\begin{aligned}\overline{\theta}_c &= \frac{1}{N_c}\sum_{\ell\in\mathcal{L}_c} n_{c,\ell}\theta_{c,\ell} \\ &= \beta^G + \beta_c^C + \frac{1}{N_c}\sum_{\ell\in\mathcal{L}_c} n_{c,\ell}\beta_{c,\ell}^L. \end{aligned}\tag{Appendix A.88}$$

The final term vanishes, so

$$\beta_c^C = \overline{\theta}_c - \beta^G. \tag{Appendix A.89}$$

Finally,

$$\beta_{c,\ell}^L = \theta_{c,\ell} - \beta^G - \beta_c^C = \theta_{c,\ell} - \overline{\theta}_c. \tag{Appendix A.90}$$

Every component is therefore uniquely determined. □

Appendix A.15. Reduced identifiable coordinates

Let $\boldsymbol{w}_C \in \mathbb{R}^C$ contain the positive chemistry weights N_c. An orthonormal null-space matrix $\boldsymbol{N}_C \in \mathbb{R}^{C\times(C-1)}$ satisfies

$$\boldsymbol{w}_C^\top \boldsymbol{N}_C = \boldsymbol{0}^\top, \qquad \boldsymbol{N}_C^\top \boldsymbol{N}_C = \boldsymbol{I}. \tag{Appendix A.91}$$

Every chemistry-deviation vector satisfying the weighted zero-sum constraint can be represented as

$$\boldsymbol{\beta}^C = \boldsymbol{N}_C \boldsymbol{\gamma}^C. \tag{Appendix A.92}$$

For chemistry c, let $\boldsymbol{w}_{L,c}$ contain the positive cell weights. An orthonormal null-space matrix satisfies

$$\boldsymbol{w}_{L,c}^\top \boldsymbol{N}_{L,c} = \boldsymbol{0}^\top, \qquad \boldsymbol{N}_{L,c}^\top \boldsymbol{N}_{L,c} = \boldsymbol{I}. \tag{Appendix A.93}$$

The cell deviations are

$$\boldsymbol{\beta}_c^L = \boldsymbol{N}_{L,c} \boldsymbol{\gamma}_c^L. \tag{Appendix A.94}$$

Collecting all transformed channels and basis terms gives

$$\boldsymbol{\beta}^{\text{full}} = \boldsymbol{A}_{\text{id}} \boldsymbol{\beta}^{\text{red}}, \tag{Appendix A.95}$$

where $\boldsymbol{A}_{\text{id}}$ is fixed before fitting.

For one transformed channel and one basis term,

$$\begin{aligned} p_{\text{one}} &= 1 + (C-1) + \sum_{c=1}^{C} (|\mathcal{L}_c| - 1) \\ &= 1 + (C-1) + (L-C) \\ &= L. \end{aligned} \tag{Appendix A.96}$$

With $C = 3$ chemistries and $L = 8$ cells,

$$p_{\text{one}} = 8. \tag{Appendix A.97}$$

With five transformed channels and six basis terms,

$$p_{\text{red}} = 5 \times 6 \times 8 = 240. \tag{Appendix A.98}$$

The nominal unconstrained full dimension is

$$p_{\text{full}} = 5 \times 6 \times (1 + 3 + 8) = 360. \tag{Appendix A.99}$$

Appendix A.16. Candidate-family definitions

The candidate set is

$$\mathcal{M} = \{M_0, M_1, M_2, M_3, M_4\}. \qquad \text{(Appendix A.100)}$$

Each candidate is represented by a reduced-coordinate mask

$$\boldsymbol{m}_M^{\text{cand}} \in \{0,1\}^{240}. \qquad \text{(Appendix A.101)}$$

The candidate-enabled vector is

$$\boldsymbol{\beta}_M = \boldsymbol{m}_M^{\text{cand}} \odot \boldsymbol{\beta}^{\text{red}}. \qquad \text{(Appendix A.102)}$$

Appendix A.16.1. Candidate M0: fixed 2RC model

Only the five global intercepts are enabled:

$$\eta_{r,s,k} = \beta_{r,1}^{G}, \qquad r \in \mathcal{R}, \qquad \text{(Appendix A.103)}$$

so

$$p_{M_0} = 5. \qquad \text{(Appendix A.104)}$$

Appendix A.16.2. Candidate M1: global dense LPV model

All six global basis terms are enabled:

$$\eta_{r,s,k} = \sum_{m=1}^{6} \beta_{r,m}^{G} \phi_m(\boldsymbol{\xi}_{s,k}), \qquad \text{(Appendix A.105)}$$

and

$$p_{M_1} = 5 \times 6 = 30. \qquad \text{(Appendix A.106)}$$

Appendix A.16.3. Candidate M2: global sparse LPV model

Candidate M_2 has the same enabled coefficient space as M_1:

$$\mathcal{A}_{M_2} = \mathcal{A}_{M_1}, \qquad p_{M_2} = 30, \qquad \text{(Appendix A.107)}$$

but applies group-sparsity regularization to the nonconstant global basis terms. There are five nonconstant basis terms for each of five transformed channels:

$$|\mathcal{G}_{M_2}| = 5 \times 5 = 25. \qquad \text{(Appendix A.108)}$$

Appendix A.16.4. Candidate M3: hierarchical dense LPV model

Candidate M_3 enables the complete identifiable hierarchy:

$$\eta_{r,s,k} = \sum_{m=1}^{6} \left(\beta^{G}_{r,m} + \beta^{C}_{r,c_s,m} + \beta^{L}_{r,c_s,\ell_s,m}\right) \phi_m(\boldsymbol{\xi}_{s,k}), \qquad \text{(Appendix A.109)}$$

with

$$p^{\text{full}}_{M_3} = 360, \qquad p^{\text{red}}_{M_3} = 240. \qquad \text{(Appendix A.110)}$$

Appendix A.16.5. Candidate M4: hierarchical sparse LPV model

Candidate M_4 has the same enabled identifiable space as M_3:

$$\mathcal{A}_{M_4} = \mathcal{A}_{M_3}, \qquad p^{\text{red}}_{M_4} = 240. \qquad \text{(Appendix A.111)}$$

The sparse group set contains

$$25$$

nonconstant global groups,

$$30$$

chemistry-deviation groups, and

$$90$$

cell-deviation groups. Hence,

$$|\mathcal{G}_{M_4}| = 25 + 30 + 90 = 145. \qquad \text{(Appendix A.112)}$$

Appendix A.17. Nestedness of the candidate spaces

The coefficient spaces satisfy

$$\mathcal{A}_{M_0} \subset \mathcal{A}_{M_1} = \mathcal{A}_{M_2} \subset \mathcal{A}_{M_3} = \mathcal{A}_{M_4}. \qquad \text{(Appendix A.113)}$$

Proof.. Candidate M_0 is obtained from M_1 by setting all nonconstant global coefficients to zero. Candidates M_1 and M_2 enable the same global space and differ only in regularization. Candidate M_1 is obtained from M_3 by setting all chemistry- and cell-level deviations to zero. Candidates M_3 and M_4 enable the same hierarchical space and differ only in regularization. □

Appendix A.18. Initialization from the fixed 2RC benchmark

Every LPV candidate is initialized from the fixed 2RC parameters:

$$\eta_{R_i}^{(0)} = \log R_i^{\text{fixed}}, \qquad i \in \{0, 1, 2\}, \tag{Appendix A.114}$$

$$\eta_{\tau_1}^{(0)} = \log \tau_1^{\text{fixed}}, \tag{Appendix A.115}$$

and

$$\eta_{\Delta\tau}^{(0)} = \log\left(\tau_2^{\text{fixed}} - \tau_1^{\text{fixed}}\right). \tag{Appendix A.116}$$

These values initialize the global intercepts. All nonconstant, chemistry-level, and cell-level deviations begin at zero.

Appendix A.19. Residual and robust data loss

The signed residual is

$$e_{s,k} = V_{s,k} - \widehat{V}_{s,k}. \tag{Appendix A.117}$$

Thus, $e_{s,k} < 0$ denotes voltage overprediction and $e_{s,k} > 0$ denotes voltage underprediction.

Let

$$m_{s,k}^{\text{loss}} \in \{0, 1\} \tag{Appendix A.118}$$

denote the valid loss mask, with

$$m_{s,0}^{\text{loss}} = 0. \tag{Appendix A.119}$$

The robust loss follows the Huber robust-estimation principle [**?**]. The pseudo-Huber loss is

$$\ell_{\text{PH}}(e; \delta) = \delta^2 \left[\sqrt{1 + \left(\frac{e}{\delta}\right)^2} - 1\right], \qquad \delta = 0.020 \text{ V}. \tag{Appendix A.120}$$

Its first derivative is

$$\frac{\partial \ell_{\text{PH}}}{\partial e} = \frac{e}{\sqrt{1 + (e/\delta)^2}}, \tag{Appendix A.121}$$

and its second derivative is

$$\frac{\partial^2 \ell_{\text{PH}}}{\partial e^2} = \frac{1}{\left[1 + (e/\delta)^2\right]^{3/2}} > 0. \tag{Appendix A.122}$$

The scalar loss is smooth and strictly convex in e, although the complete recursive model remains nonlinear and nonconvex in its coefficient vector.

Let

$$N_{\text{loss}} = \sum_{s,k} m_{s,k}^{\text{loss}}. \qquad \text{(Appendix A.123)}$$

The normalized data term is

$$\mathcal{L}_{\text{data}} = \frac{\sum_{s,k} m_{s,k}^{\text{loss}} \ell_{\text{PH}}\left(e_{s,k};\delta\right)}{\max(N_{\text{loss}},1)}. \qquad \text{(Appendix A.124)}$$

Appendix A.20. Ridge, group-sparsity, and physical-range penalties

The ridge term is

$$\mathcal{L}_{\text{ridge},M} = \lambda_{\text{ridge}} \frac{\left\|\boldsymbol{m}_M^{\text{cand}} \odot \boldsymbol{\beta}^{\text{red}}\right\|_2^2}{240}, \qquad \lambda_{\text{ridge}} = 10^{-7}. \qquad \text{(Appendix A.125)}$$

Grouped coefficient selection follows the group-lasso principle [?]. For $M \in \{M_2, M_4\}$, the group-sparsity penalty is

$$\mathcal{P}_M(\boldsymbol{\beta}) = \sum_{g\in\mathcal{G}_M} \sqrt{|g|}\,\|\boldsymbol{\beta}_g\|_2\,. \qquad \text{(Appendix A.126)}$$

For a nonzero group,

$$\nabla_{\boldsymbol{\beta}_g}\left[\sqrt{|g|}\,\|\boldsymbol{\beta}_g\|_2\right] = \sqrt{|g|}\frac{\boldsymbol{\beta}_g}{\|\boldsymbol{\beta}_g\|_2}. \qquad \text{(Appendix A.127)}$$

At $\boldsymbol{\beta}_g = \mathbf{0}$, the subdifferential is

$$\partial\left[\sqrt{|g|}\|\boldsymbol{\beta}_g\|_2\right] = \left\{\boldsymbol{u} : \|\boldsymbol{u}\|_2 \le \sqrt{|g|}\right\}. \qquad \text{(Appendix A.128)}$$

The implementation uses the zero subgradient.

Define

$$d_{s,k,r}^{-} = \max\left(\eta_r^{\min} - \eta_{r,s,k}, 0\right), \qquad \text{(Appendix A.129)}$$

and

$$d_{s,k,r}^{+} = \max\left(\eta_{r,s,k} - \eta_r^{\max}, 0\right). \qquad \text{(Appendix A.130)}$$

The physical-range penalty is

$$\mathcal{B}_{\text{phys}} = \frac{\sum_{s,k} m_{s,k}^{\text{loss}} \sum_{r\in\mathcal{R}} \left[\left(d_{s,k,r}^{-}\right)^2 + \left(d_{s,k,r}^{+}\right)^2\right]}{5\max(N_{\text{loss}},1)}. \qquad \text{(Appendix A.131)}$$

The complete objective is

$$\mathcal{J}_M = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{ridge},M} + \lambda_M \mathcal{P}_M + \lambda_{\text{phys}}\mathcal{B}_{\text{phys}}, \qquad \text{(Appendix A.132)}$$

with

$$\lambda_{\text{phys}} = 10^{-3}. \qquad \text{(Appendix A.133)}$$

For dense candidates,

$$\lambda_M = 0, \qquad M \in \{M_0, M_1, M_3\}, \qquad \text{(Appendix A.134)}$$

whereas sparse candidates use

$$\lambda_M \in \left\{10^{-6}, 3\times10^{-6}, 10^{-5}, 3\times10^{-5}, 10^{-4}\right\}. \qquad \text{(Appendix A.135)}$$

Appendix A.21. Optimization configuration

The candidate jobs are optimized with Adam [?]. Adam uses

$$\alpha_{\text{Adam}} = 2\times10^{-3}, \qquad \beta_{1,\text{Adam}} = 0.9, \qquad \beta_{2,\text{Adam}} = 0.999, \qquad \epsilon_{\text{Adam}} = 10^{-8}. \qquad \text{(Appendix A.136)}$$

The trajectory-batch size is

$$B_{\text{traj}} = 8. \qquad \text{(Appendix A.137)}$$

For the raw gradient $\boldsymbol{g}^{(u)}$ at optimization iteration u,

$$\widetilde{\boldsymbol{g}}^{(u)} = \boldsymbol{g}^{(u)} \min\left(1, \frac{10}{\|\boldsymbol{g}^{(u)}\|_2}\right), \qquad \text{(Appendix A.138)}$$

so

$$\left\|\widetilde{\boldsymbol{g}}^{(u)}\right\|_2 \le 10. \qquad \text{(Appendix A.139)}$$

Dense jobs use 1800 update steps, sparse jobs use 2400 update steps, and the selected-support refit uses 1200 update steps.

Appendix A.22. Support thresholding and heredity closure

A scalar coefficient is active when

$$|\beta_i| > \epsilon_{\text{support}}, \tag{Appendix A.140}$$

and a sparse group is active when

$$\|\boldsymbol{\beta}_g\|_2 > \epsilon_{\text{support}}, \tag{Appendix A.141}$$

where

$$\epsilon_{\text{support}} = 10^{-4}. \tag{Appendix A.142}$$

The closure rule applies a strong-heredity interpretation to polynomial main effects and interactions [7]. Basis heredity requires

$$\xi_z^2 \Longrightarrow \xi_z, \tag{Appendix A.143}$$

$$\xi_T^2 \Longrightarrow \xi_T, \tag{Appendix A.144}$$

and

$$\xi_z \xi_T \Longrightarrow \{\xi_z, \xi_T\}. \tag{Appendix A.145}$$

Hierarchy heredity requires

$$\text{CELL} \Longrightarrow \text{CHEMISTRY} \Longrightarrow \text{GLOBAL}. \tag{Appendix A.146}$$

Hierarchy closure is applied within the same transformed channel and basis term: an active cell-level coefficient requires its corresponding chemistry-level and global parent coefficients.

Let $\mathcal{A}_{\text{raw}}$ denote the thresholded support and let

$$\text{cl}\left(\mathcal{A}_{\text{raw}}\right) \tag{Appendix A.147}$$

denote the smallest support containing $\mathcal{A}_{\text{raw}}$ and satisfying all heredity rules.

Proposition A.2.. The heredity-closure operator is extensive, monotone, idempotent, and minimal.

Proof.. It is extensive because

$$\mathcal{A} \subseteq \mathrm{cl}(\mathcal{A}).$$

It is monotone because

$$\mathcal{A} \subseteq \mathcal{B} \quad \Longrightarrow \quad \mathrm{cl}(\mathcal{A}) \subseteq \mathrm{cl}(\mathcal{B}).$$

It is idempotent because the first closure already satisfies all heredity rules:

$$\mathrm{cl}\left(\mathrm{cl}(\mathcal{A})\right) = \mathrm{cl}(\mathcal{A}).$$

The coefficient dictionary is finite, so repeatedly adding required parent terms terminates after finitely many additions. The resulting set is contained in every heredity-valid superset of $\mathcal{A}$, which proves minimality. □

Let

$$\boldsymbol{m}^{\mathrm{closed}} \in \{0,1\}^{240} \tag{Appendix A.148}$$

denote the closed support mask.

Appendix A.23. Training-only selected-support refit

After candidate selection and heredity closure, coefficients outside the closed support are fixed to zero. The refit objective is

$$\mathcal{J}_{\mathrm{refit}} = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{ridge}} \frac{\left\| \boldsymbol{m}^{\mathrm{closed}} \odot \boldsymbol{\beta}^{\mathrm{red}} \right\|_2^2}{240} + \lambda_{\mathrm{phys}} \mathcal{B}_{\mathrm{phys}}. \tag{Appendix A.149}$$

The group-sparsity term is not used during refitting. Structural selection has already been completed, so the refit estimates the coefficients of the closed structure without continuing group shrinkage.

Appendix A.24. Calibration-stage candidate selection

For candidate job j, let $\mathcal{S}_{\mathrm{cal}}$ denote the calibration trajectories with at least one valid evaluated sample, and let $\mathcal{K}_s^{\mathrm{eval}}$ denote the valid evaluated sample indices of trajectory s. Define

$$N_s^{\mathrm{eval}} = \left| \mathcal{K}_s^{\mathrm{eval}} \right|. \tag{Appendix A.150}$$

The calibration trajectory RMSE is

$$\mathrm{RMSE}_{j,s}^{\mathrm{cal}} = \sqrt{\frac{1}{N_s^{\mathrm{eval}}} \sum_{k \in \mathcal{K}_s^{\mathrm{eval}}} e_{j,s,k}^2}. \tag{Appendix A.151}$$

The trajectory-weighted calibration score is

$$E_j^{\mathrm{cal}} = \frac{1}{|\mathcal{S}_{\mathrm{cal}}|} \sum_{s \in \mathcal{S}_{\mathrm{cal}}} \mathrm{RMSE}_{j,s}^{\mathrm{cal}}\,. \tag{Appendix A.152}$$

Each calibration trajectory therefore receives equal weight.

Let

$$E_{\min}^{\mathrm{cal}} = \min_j E_j^{\mathrm{cal}}, \tag{Appendix A.153}$$

and

$$\epsilon_{\mathrm{cal,eq}} = 0.0025. \tag{Appendix A.154}$$

The calibration-equivalent pool is

$$\mathcal{A}_{\mathrm{cal}} = \left\{ j : E_j^{\mathrm{cal}} \le (1 + \epsilon_{\mathrm{cal,eq}})\, E_{\min}^{\mathrm{cal}} \right\}. \tag{Appendix A.155}$$

Jobs in $\mathcal{A}_{\mathrm{cal}}$ are ordered lexicographically by active group count, nonzero coefficient count, active candidate coordinate count, trajectory-weighted calibration RMSE, candidate identifier, regularization value, and job identifier. The first job is selected.

Appendix A.25. Model freezing and test isolation

Before ordinary-test or strict-test evaluation, the following objects are frozen:

$$\mathcal{F} = \begin{Bmatrix} M^\star, \lambda^\star, \boldsymbol{m}^{\mathrm{closed}}, \boldsymbol{\beta}^\star, \boldsymbol{\chi}_{\mathrm{mid}}, \boldsymbol{\chi}_{\mathrm{half}}, \\ \boldsymbol{A}_{\mathrm{id}}, \boldsymbol{\eta}^{\min}, \boldsymbol{\eta}^{\max},\ \text{initial-state rule},\ \text{simulation configuration} \end{Bmatrix}. \tag{Appendix A.156}$$

Proposition A.3.. Under the implemented sequence, ordinary-test and strict-test observations cannot affect candidate identity, regularization selection, support selection, refitted coefficients, or scheduler normalization. The training partition determines scheduler normalization, hierarchy weights, identifiable expansion, fitted candidate coefficients, and the selected-support refit. The calibration partition determines the candidate job and regularization choice. The frozen set $\mathcal{F}$ is completed before either test partition is opened. Ordinary-test and strict-test data are subsequently passed only to the fixed prediction and diagnostic functions. No component of $\mathcal{F}$ is recomputed after test access. □

This is a procedural test-isolation result. It does not imply that the partitions are identically distributed, nor does it eliminate distribution shift.

Appendix A.26. Accuracy metrics

Let

$$\mathcal{E} = \left\{(s,k) : k \geq 1,\ m^{\text{eval}}_{s,k} = 1\right\} \tag{Appendix A.157}$$

denote a nonempty pooled evaluation set. The pooled RMSE is

$$\text{RMSE}_{\text{pool}} = \sqrt{\frac{1}{|\mathcal{E}|} \sum_{(s,k)\in\mathcal{E}} e^2_{s,k}}. \tag{Appendix A.158}$$

The pooled MAE is

$$\text{MAE}_{\text{pool}} = \frac{1}{|\mathcal{E}|} \sum_{(s,k)\in\mathcal{E}} |e_{s,k}|. \tag{Appendix A.159}$$

The pooled bias is

$$\text{Bias}_{\text{pool}} = \frac{1}{|\mathcal{E}|} \sum_{(s,k)\in\mathcal{E}} e_{s,k}. \tag{Appendix A.160}$$

For trajectory s,

$$\text{RMSE}_s = \sqrt{\frac{1}{N^{\text{eval}}_s} \sum_{k\in\mathcal{K}^{\text{eval}}_s} e^2_{s,k}}. \tag{Appendix A.161}$$

The trajectory-weighted RMSE is

$$\text{RMSE}_{\text{TW}} = \frac{1}{|\mathcal{S}_{\text{eval}}|} \sum_{s\in\mathcal{S}_{\text{eval}}} \text{RMSE}_s\,. \tag{Appendix A.162}$$

The pooled RMSE weights every sample equally, whereas the trajectory-weighted RMSE weights every trajectory equally.

Appendix A.27. Residual-dependence diagnostics

For diagnostic variable $x_{s,k}$, the sample-weighted Pearson correlation is

$$\rho(e,x) = \frac{\sum_{(s,k)\in\mathcal{E}} (e_{s,k} - \overline{e})\,(x_{s,k} - \overline{x})}{\sqrt{\sum_{(s,k)\in\mathcal{E}} (e_{s,k} - \overline{e})^2}\sqrt{\sum_{(s,k)\in\mathcal{E}} (x_{s,k} - \overline{x})^2}}. \tag{Appendix A.163}$$

This statistic is defined when both variables have positive sample variance. It quantifies linear association and does not establish general independence or causality.

For diagnostic bin $\mathcal{B}_b$,

$$\overline{e}_b = \frac{\sum_{(s,k)\in\mathcal{E}} \mathbb{I}\left\{x_{s,k} \in \mathcal{B}_b\right\} e_{s,k}}{\sum_{(s,k)\in\mathcal{E}} \mathbb{I}\left\{x_{s,k} \in \mathcal{B}_b\right\}}. \tag{Appendix A.164}$$

Equation (Appendix A.164) is evaluated only for bins with a positive sample count.

For SOC bin $\mathcal{Z}_a$ and temperature bin $\mathcal{T}_b$,

$$n_{a,b} = \sum_{(s,k)\in\mathcal{E}} \mathbb{I}\left\{z_{s,k} \in \mathcal{Z}_a\right\} \mathbb{I}\left\{T_{s,k} \in \mathcal{T}_b\right\}, \tag{Appendix A.165}$$

and

$$\overline{e}_{a,b} = \frac{\sum_{(s,k)\in\mathcal{E}} \mathbb{I}\left\{z_{s,k} \in \mathcal{Z}_a\right\} \mathbb{I}\left\{T_{s,k} \in \mathcal{T}_b\right\} e_{s,k}}{n_{a,b}}. \tag{Appendix A.166}$$

The joint mean is interpreted only when $n_{a,b} > 0$, together with the minimum-count rule used in the reported diagnostic figures.

The training scheduler-support count is

$$n^{\mathrm{tr}}_{a,b} = \sum_{(s,k)\in\mathcal{E}_{\mathrm{tr}}} \mathbb{I}\left\{\xi_{z,s,k} \in \Xi_a\right\} \mathbb{I}\left\{\xi_{T,s,k} \in \Theta_b\right\}. \tag{Appendix A.167}$$

This describes sample density in the fitted SOC–temperature plane; it does not establish chemistry-, cell-, current-history-, or trajectory-level support.

Appendix A.28. Centered mini-batch gradient-span diagnostic

Let $p_{\mathcal{S}}$ be the number of selected reduced coefficients, and let

$$\boldsymbol{g}_b = \nabla_{\boldsymbol{\beta}_{\mathcal{S}}} \mathcal{L}^{(b)}_{\mathrm{data}}\left(\boldsymbol{\beta}^{\star}\right) \in \mathbb{R}^{p_{\mathcal{S}}} \tag{Appendix A.168}$$

be the selected-support data-loss gradient for training mini-batch b. Stack B gradients into

$$\boldsymbol{G} = \begin{bmatrix} \boldsymbol{g}_1^{\top} \\ \vdots \\ \boldsymbol{g}_B^{\top} \end{bmatrix} \in \mathbb{R}^{B\times p_{\mathcal{S}}}. \tag{Appendix A.169}$$

The mean gradient is

$$\overline{\boldsymbol{g}} = \frac{1}{B} \sum_{b=1}^{B} \boldsymbol{g}_b, \tag{Appendix A.170}$$

and the centered matrix is

$$\boldsymbol{G}_c = \boldsymbol{G} - \mathbf{1}_B \overline{\boldsymbol{g}}^\top. \tag{Appendix A.171}$$

Equivalently,

$$\boldsymbol{G}_c = \boldsymbol{H}_B \boldsymbol{G}, \qquad \boldsymbol{H}_B = \boldsymbol{I}_B - \frac{1}{B} \mathbf{1}_B \mathbf{1}_B^\top. \tag{Appendix A.172}$$

Since

$$\operatorname{rank}(\boldsymbol{H}_B) = B - 1, \tag{Appendix A.173}$$

$$\operatorname{rank}(\boldsymbol{G}_c) \le \min(B - 1, p_{\mathcal{S}}). \tag{Appendix A.174}$$

The singular-value analysis follows standard numerical linear-algebra practice [8].

Let

$$\boldsymbol{G}_c = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^\top \tag{Appendix A.175}$$

be the singular-value decomposition. The reported practical gradient-span rank is

$$r_{\text{grad}} = \# \left\{ i : \sigma_i(\boldsymbol{G}_c) > \max(B, p_{\mathcal{S}}) \epsilon_{\text{mach}} \sigma_1(\boldsymbol{G}_c) \right\}. \tag{Appendix A.176}$$

This quantity measures the numerical span of centered mini-batch gradient directions at the selected solution. It is not the rank of the complete recursive residual Jacobian, not a Fisher-information rank, and not a full parameter-identifiability proof.

References

[1] S. Nejad, D. T. Gladwin, and D. A. Stone, "A systematic review of lumped-parameter equivalent circuit models for real-time estimation of lithium-ion battery states," *Journal of Power Sources*, vol. 316, pp. 183–196, 2016. doi: 10.1016/j.jpowsour.2016.03.042.

[2] X. Hu, S. Li, and H. Peng, “A comparative study of equivalent circuit models for Li-ion batteries,” *Journal of Power Sources*, vol. 198, pp. 359–367, 2012. doi: 10.1016/j.jpowsour.2011.10.013.

[3] A. Nikolian, J. Jaguemont, J. de Hoog, S. Goutam, N. Omar, P. Van Den Bossche, and J. Van Mierlo, “Complete cell-level lithium-ion electrical ECM model for different chemistries (NMC, LFP, LTO) and temperatures ($-5\,^{\circ}$C to $45\,^{\circ}$C)—optimized modelling techniques,” *International Journal of Electrical Power & Energy Systems*, vol. 98, pp. 133–146, 2018. doi: 10.1016/j.ijepes.2017.11.031.

[4] Y. Sun et al., “A state-dependent quasi-linear parameter-varying model of lithium-ion batteries for state of charge estimation,” *Journal of Power Sources*, vol. 614, Art. no. 234879, 2024. doi: 10.1016/j.jpowsour.2024.234879.

[5] Z. Huang, M. C. Best, J. Knowles, and A. Fly, “Adaptive piecewise equivalent circuit model with SOC/SOH estimation based on extended Kalman filter,” *IEEE Transactions on Energy Conversion*, vol. 38, no. 2, pp. 959–970, 2023. doi: 10.1109/TEC.2022.3218613.

[6] A. M. A. Sheikh, M. C. F. Donkers, and H. J. Bergveld, “A comprehensive approach to sparse identification of linear parameter-varying models for lithium-ion batteries using improved experimental design,” *Journal of Energy Storage*, vol. 95, Art. no. 112581, 2024. doi: 10.1016/j.est.2024.112581.

[7] J. Bien, J. Taylor, and R. Tibshirani, “A lasso for hierarchical interactions,” *The Annals of Statistics*, vol. 41, no. 3, pp. 1111–1141, 2013. doi: 10.1214/13-AOS1096.

[8] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed. Baltimore, MD, USA: Johns Hopkins University Press, 2013.